

Bearing Degradation Prediction based on Multi-Scale Mamba-Transformer Model

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Abstract—Rolling bearings are critical components of rotating machinery, directly impacting equipment reliability and service life. Accurately predicting bearing degradation trends is essential for ensuring system safety and stable operation. However, the degradation process of bearings exhibits complex multi-stage characteristics, with local non-stationarities and irregular fluctuations, creating challenges in modeling due to the heterogeneity between global and local dynamics. To address this issue, we proposed a Multi-Scale Mamba-Transformer Model (MSMT) and employed it for bearing degradation prediction. The MSMT model leverages the Mamba architecture to capture long-term degradation trends while incorporating a local attention mechanism to extract short-term deterioration features, thereby enhancing both prediction accuracy and robustness. To validate the effectiveness of the proposed model, experiments are conducted on the PRONOSTIA bearing degradation dataset, comparing MSMT with multiple existing methods. The experimental results demonstrate that the proposed model significantly outperforms baseline approaches in prediction accuracy, highlighting its superior performance in bearing degradation trend forecasting.

Keywords—bearings, degradation prediction, Mamba, Attention Mechanism

I. INTRODUCTION

Prognostics and Health Management (PHM) is of paramount importance in ensuring the safety, reliability, and efficient operation of mechanical equipment [1]. Among various mechanical systems, rolling bearings, as critical components in rotating machinery, are prone to

faults such as pitting, spalling, and wear, due to prolonged operation under high loads and varying working conditions[2]. Therefore, accurately predicting the degradation trend of rolling bearings not only facilitates timely maintenance actions to prevent catastrophic failures but also significantly reduces the economic losses resulting from equipment downtime [3]. This critical task has garnered widespread attention in the field of PHM, carrying significant theoretical research value and practical application significance.

In recent years, regarding the prediction of bearing degradation trends, researchers have primarily adopted two major categories of approaches: model-driven methods and data-driven methods[4]. Model-driven approaches emphasize the development of mathematical models for degradation mechanisms based on physical and mathematical theories; however, in practical industrial environments—particularly when dealing with complex machinery—accurately establishing such models remains a significant challenge [5]. In contrast, with the rapid advancements in artificial intelligence and big data analytics, data-driven methods have emerged as a prominent research focus. These approaches utilize equipment condition monitoring data in conjunction with advanced machine learning algorithms to predict degradation trends, thereby effectively compensating for the limitations associated with model-driven methods in real-world applications[6].

In data-driven approaches, deep learning has demonstrated immense potential in degradation trend prediction, owing to its hierarchical feature extraction capabilities. Its multi-layered architecture not only

adeptly extracts latent patterns from complex datasets but also captures long-term dependencies, thereby enhancing predictive accuracy. Representative techniques such as Recurrent Neural Networks (RNNs) and Transformers have emerged as powerful tools for degradation modeling. For instance, Su et al. [7] employed a Stacked Autoencoder (SAE) to extract signal features, which were subsequently input into a Long Short-Term Memory (LSTM) network for prediction; Yan et al. [8] constructed a prediction model based on Gated Recurrent Units (GRUs) using degradation indices as input. Hao et al. [9] leveraged the Transformer to extract deep features that encapsulate richer degradation process information from the input samples and implemented life prediction within a classical RUL prediction network comprising multiple fully connected layers. However, given that vibration signals inherently exhibit nonlinear and non-stationary characteristics, a single deep learning model still faces limitations in capturing intricate degradation patterns and suppressing redundant data, which in turn adversely affects its generalization capability and predictive accuracy. Consequently, enhancing the representation of local features while concurrently modeling global dynamics remains a pivotal challenge in advancing the accuracy of degradation predictions.

To address the aforementioned issues, this paper proposes a bearing degradation trend prediction method based on Multi-Scale Mamba-Transformer (MSMT). The method aims to achieve comprehensive modeling of long-range global information and short-term local dynamic features by integrating the global MAMBA model with a local attention mechanism. Specifically, the global modeling module leverages the MAMBA framework to extract the overall degradation trend and long-term dependencies, while the local degradation modeling module employs a sliding window technique to segment the raw time series into multiple local subsequences. These subsequences are then processed using a Transformer-based local encoder to capture fine-grained degradation information. The effective fusion of global and local features not only provides a comprehensive representation of multi-scale information in the bearing degradation process but also overcomes the limitations of a single model in feature learning, significantly improving both prediction accuracy and robustness. Extensive experiments conducted on the PHM 2012 dataset demonstrate that the proposed method outperforms existing state-of-the-art approaches in terms of prediction accuracy and generalization performance.

II. METHODOLOGY

A. Global Degradation Module Based on MAMBA

A critical challenge in bearing degradation prediction is the inherently long-term and intricate nature of the

degradation process. Existing methodologies aimed at capturing long-range dependencies and complex contextual information typically suffer from two primary limitations. On one hand, certain approaches employ fixed-size contextual windows (e.g., attention-based mechanisms), which inherently constrain the range of information captured and may consequently result in the omission of vital features. On the other hand, some methods rely on heuristic strategies with parameter adjustments based on empirical rules, thereby exhibiting a limited memory capacity. For example, recurrent neural networks (RNNs) often encounter difficulties in adequately retaining and propagating essential information when processing extended sequences.

To address these challenges, we propose a long-term degradation trend modeling network based on the Mamba model [10]. This network incorporates a Selectivity Mechanism into the State Space Model (SSM), thereby developing a Selective State Space Model that can dynamically adjust its parameters in response to varying input data patterns.

The State Space Model, as illustrated in Figure 1, is a widely used tool for time-series analysis, providing a comprehensive and versatile framework for modeling sequential data. These models not only represent the internal states of a system but also elucidate the relationships between these states and external input-output variables. A conventional SSM consists of two fundamental equations: the state transition equation and the observation equation. The state transition equation characterizes the internal dynamics of the system and describes how these states evolve over time under the influence of input variables, serving as the core mechanism governing system changes. Meanwhile, the observation equation provides the system's state at a given time, reflecting how the system responds to variations in input variables.

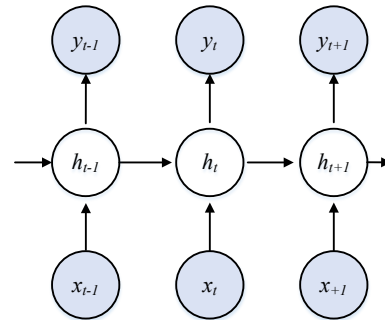


Fig. 1. State Space Model

Initially, the continuous state-space model is discretized using the zero-order hold (ZOH) method, thereby effectively transitioning to a discrete model as shown in (1-2). The ZOH approach maintains a constant input over each sampling interval, simplifying the

translation of the continuous system into its discrete equivalent. By preserving the most recent value until the subsequent input is sampled, the zero-order hold ensures that the system can be accurately represented in discrete time without forfeiting critical information during the conversion. This process yields the following discrete model equation:

$$h_t = \bar{A}h_{t-1} + \bar{B}x_t \quad (1)$$

$$y_t = Ch_t + Dx_t \quad (2)$$

where $\bar{A} = e^{A\Delta}$, $\bar{B} = f_B(\Delta, A, B)$ Δ is a learnable parameter employed to estimate the discrete interval, which is crucial for the precise timing of input samples.

The model is specifically designed for input data characterized by a batch size B , sequence length L , and D channels. Within the Mamba framework, the state-space model (SSM) is applied to each channel individually, thereby meticulously capturing the dynamic features inherent to each channel. More importantly, the Mamba model transforms the key matrices B , C , and A into input-dependent functions, enabling the model to adapt its internal behavior based on the unique characteristics of the input data as shown in (3-5). This adaptive mechanism not only significantly enhances the model's capacity to capture long-range dependencies but also improves its robustness and effectiveness in identifying complex degradation patterns.

$$\bar{B} = S_B(x) \quad (3)$$

$$\bar{C} = S_C(x) \quad (4)$$

$$\Delta = \tau_A(\text{Parameter} + S_A(x)) \quad (5)$$

where $B \in \mathbb{R}^{B \times L \times N}$, $C \in \mathbb{R}^{B \times L \times N}$, $\Delta \in \mathbb{R}^{B \times L \times N}$. The functions $S_B(x)$ and $S_C(x)$ linearly project the input x into an N -dimensional space, whereas $S_A(x)$ projects the hidden state dimension D into the desired dimension, interfacing with the gating mechanism of recurrent neural networks (RNNs).

These computations convert the parameters A , B , and C into input-dependent functions with a length of L , thereby transforming the original time-invariant model into a time-varying model and enabling selective processing of the input content, as shown in Fig 2.

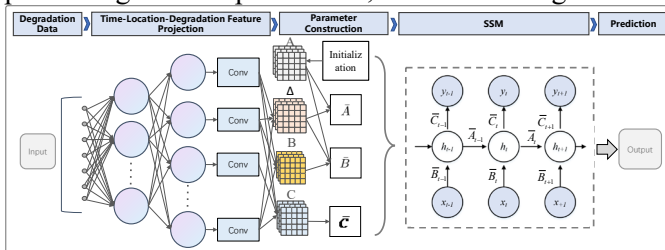


Fig. 2. Global Degradation Module Based on MAMBA

The dimension of Δ is modified from d to (b, l, d) , indicating that each token in a batch (with a total of $b \times l$ tokens) possesses a unique Δ in terms of input dependency and enhanced control functionality. A larger step size for Δ shifts the model's focus more towards the input rather than the stored state, whereas a smaller step size results in the model discounting specific input details and focusing more on the stored state. Parameters B and C have been rendered input-dependent as functions within the framework, which facilitates a more nuanced control over whether the input x influences the state h or the state h impacts the output y . Although parameter A is inherently data-independent, the discretization process in the state-space model (SSM) allows A to be affected by the input via the data dependency of Δ . Furthermore, given that parameter A has a dimension of n , it plays a distinct role across each SSM dimension, thereby enabling an accurate abstraction of all preceding content rather than a mere compression.

B. Local Degradation Module with Attention Mechanism

Due to the Mamba model's focus on long-term forecasting, it fails to selectively retain important long-term information while filtering out short-term noise. This limitation leads to a decrease in prediction accuracy when dealing with short-term fluctuations in bearing degradation data. Local variations (such as extreme values or sudden events) are more critical in short timeframes. To address this issue and capture local degradation features in bearing life prediction, this paper proposes a local degradation modeling method based on patch partitioning and a multi-head attention mechanism, as shown in Fig. 3. The method extracts local subsequences from time series using a sliding window, encodes short-term temporal features with a Time Series Transformer Encoder (TSTiEncoder), and utilizes an attention mechanism to dynamically focus on local degradation information, thus enhancing prediction accuracy.

Given a time series input $z \in \mathbb{R}^{B \times N \times T}$, where B denotes the batch size, N represents the number of variables, and T is the length of the time series, the local partitioning module employs a sliding window of length L_p with a stride of s . By applying the tensor unfolding (unfold) operation, the input sequence is segmented into multiple patches, which can be mathematically expressed as:

$$z_p = \text{unfold}(z, \text{size} = L_p, \text{step} = s) \in \mathbb{R}^{B \times N \times P \times L_p} \quad (6)$$

where $P = \left\lceil \frac{T - L_p}{s} \right\rceil + 1$ denotes the number of patches

obtained through the partitioning process.

This method ensures that each local segment thoroughly captures the dynamic variations within its corresponding time interval, thereby providing a robust foundation for the subsequent extraction of localized degradation features.

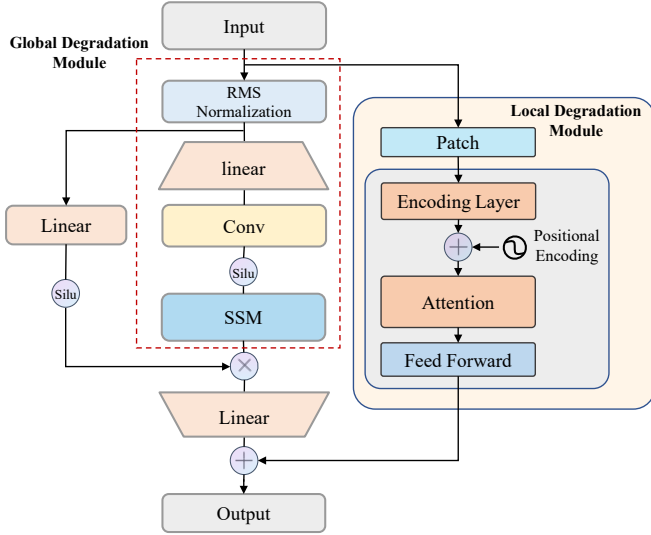


Fig. 3. Multi-Scale Mamba-Transformer Model

Considering that the input sequence length may not be an exact multiple of L_p , insufficient data may occur at the boundaries. To mitigate this issue, a replication padding strategy is employed to pad the end of the sequence, ensuring that all patches maintain a uniform length. This approach guarantees full coverage of the entire time series even in the presence of boundary effects, thereby preventing information loss or uneven feature representation.

To effectively capture the subtle features indicative of bearing degradation within the locally partitioned time series, a local feature encoder based on the Transformer architecture is constructed. This encoder performs deep encoding of the local patch data, thereby enabling efficient extraction and modeling of local temporal features. Within each local patch, a positional encoding scheme based on a combination of sine and cosine functions is incorporated to enhance the model's capacity to capture temporal dependencies. Following the linear mapping and positional encoding stages, the model further employs a multi-head attention mechanism for feature modeling to extract critical degradation patterns. In contrast to traditional global attention mechanisms, the proposed method introduces a local window strategy, computing attention distributions solely within a predefined local range to ensure the model focuses on key degradation features over short time scales. For a

fixed window size w , each token attends only to $\frac{1}{2}w$ neighboring tokens on each side. The attention scores among tokens within a window of size W are computed as shown in Equation (2):

$$Attention(Q_w, K_w, V_w) = softmax\left(\frac{Q_w K_w^T}{\sqrt{d_k}}\right) V_w \quad (7)$$

where Q_w , K_w and V_w denote query, key, and value vector within the local window, $\frac{1}{\sqrt{d_k}}$ is used to avoid extremely small gradients.

Finally, to ensure stability during training and favorable convergence of the deep network, residual connections and normalization operations are incorporated after each attention layer. The residual connections help alleviate the vanishing gradient problem, thereby enabling more stable parameter updates during optimization. Concurrently, normalization ensures consistent feature distributions across different data batches, which enhances the model's generalization capability. Ultimately, the locally processed features facilitate robust modeling of degradation data at the local level and serve as high-quality inputs for the subsequent remaining useful life (RUL) prediction module.

III. RESULTS AND DISCUSSION

A. PHM 2012 Prognostic challenge Dataset

This study utilizes the accelerated life dataset for rolling bearings provided by the IEEE PHM 2012 Challenge to validate the efficacy of the proposed model. The experimental platform is depicted in Fig. 4. Two accelerometers were mounted on the bearing's outer ring to collect vibration signals in both the horizontal and vertical directions. These signals were sampled at a frequency of 25.6 kHz, with a sampling interval of 10 s and a duration of 0.1 s per sampling session, yielding 2,560 data points every 10s[9]. A total of 17 sets of full-life data were collected from bearings operating under three different conditions; for each condition, the first two sets were used as the training set, while the remaining data were reserved for testing. Moreover, prior research indicates that horizontal vibration signals typically contain richer degradation information than vertical signals, thus this study exclusively analyzes the horizontal vibration data.

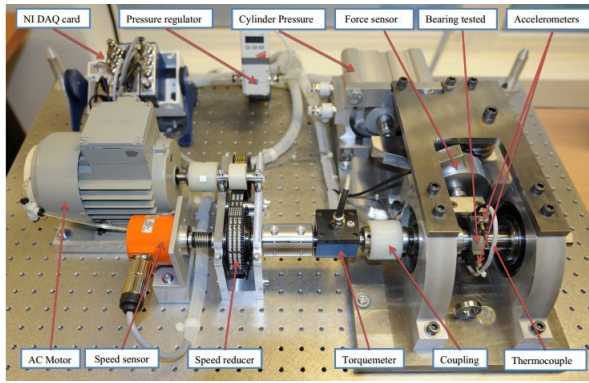


Fig. 4. PRONOSTIA experimental platform

To comprehensively characterize the degradation process of bearings, the raw data must first undergo appropriate preprocessing. In this study, 14 key indicators were selected from multiple perspectives to evaluate the health status of the bearings, including typical time-domain features such as the root mean square (RMS) and peak-to-peak value. In addition, a set of dimensionless features that are independent of operating conditions yet highly sensitive to faults and damage were chosen to enhance the model's robustness and improve its generalization capability in prediction. All selected indicators are listed in Table I.

TABLE I. FOURTEEN FEATURES TO REFLECT THE HEALTH STATUS OF BEARINGS

No.	Indicator	No.	Indicator
1	Maximum	8	kurtosis
2	Minimum	9	RMS
3	Mean	10	Waveform-factor
4	Peak-to-peak value	11	Crest-factor
5	Average absolute amplitude	12	Kurtosis-factor
6	Variance	13	Impact-factor
7	Standard	14	Clearance-factor

During the preprocessing phase, to eliminate scale differences among various physical quantities and improve data comparability, the top 10% of data points for each indicator were used to compute statistical properties. Subsequently, each indicator was standardized using (8) to obtain its standard score. Principal component analysis (PCA) was then applied to the standardized indicators for dimensionality reduction, extracting the principal components that preserve the maximum variance[11]. These principal components were subsequently used to construct a comprehensive

metric for bearing health, thereby providing a more effective description of the degradation trend. The resulting health indicator is shown in Figure 5.

$$z = \frac{x - \bar{x}}{S} \quad (8)$$

Where $\bar{x}$ denotes the mean of the first 10% of data points for each indicator, and SSS represents the standard deviation computed from the first 10% of data points for each feature.

B. Degradation Prediction Results

Based on the health indicators constructed in the previous sections, we apply the proposed method to the degraded bearing data. To better demonstrate the effectiveness of the proposed model in predicting bearing degradation trends, we compare it with three models from different domains: the classical deep learning neural network LSTM, the traditional state-space model Switching SSM, and the hybrid statistical-deep learning model DeepState.

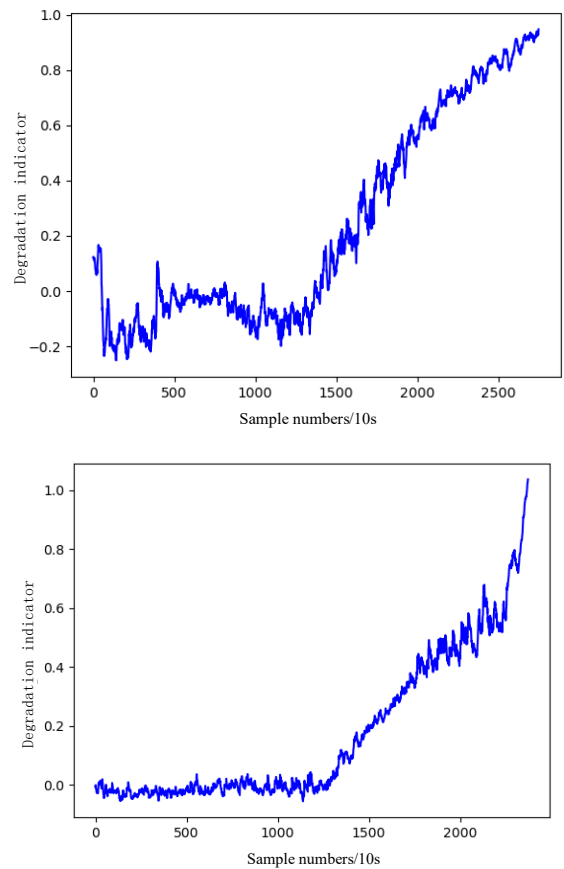


Fig. 5. Degradation indicator of Bearing1_1 and Bearing1_3

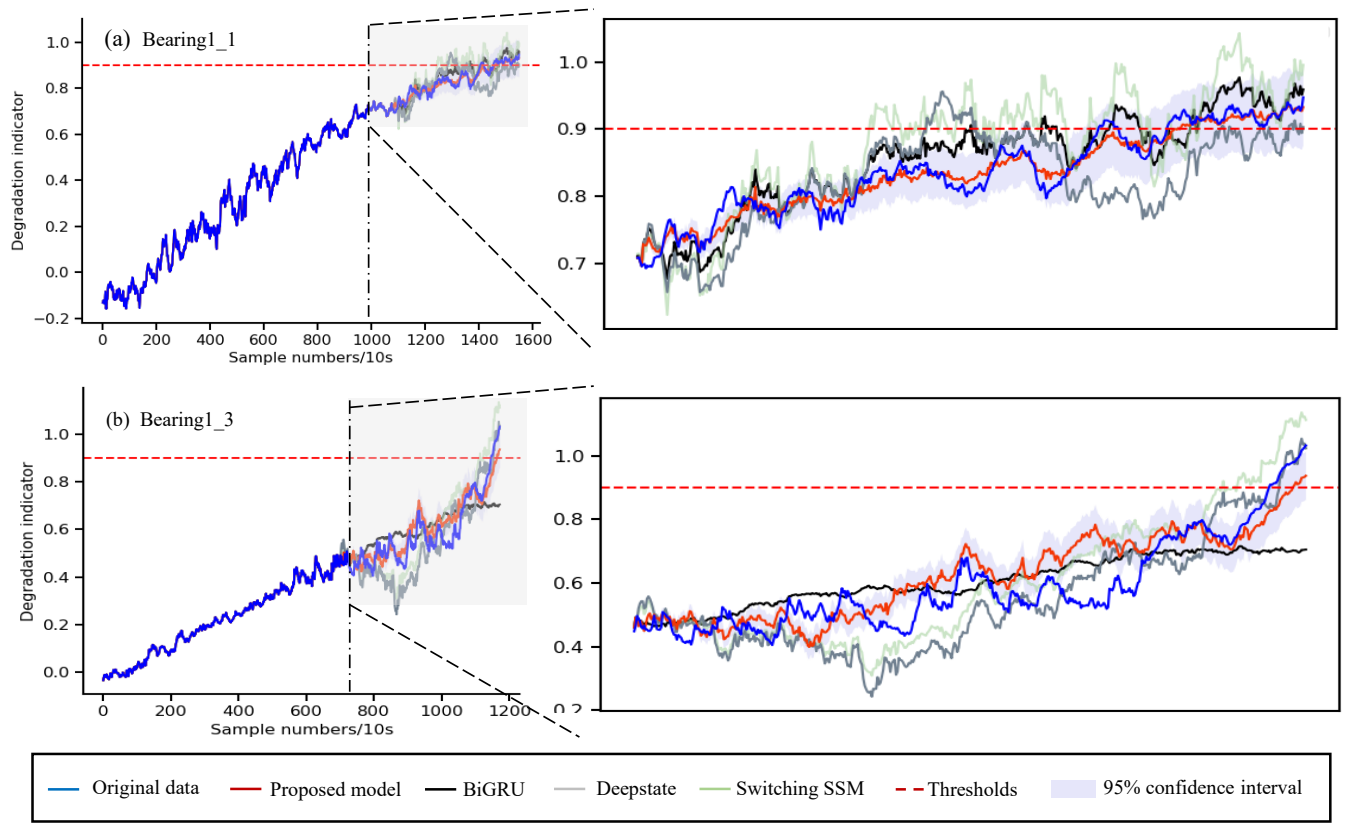


Fig. 6. Prediction results of Bearing1_1 and Bearing1_3

To quantitatively evaluate the predictive performance of bearing lifetime estimation, four metrics are employed as evaluation criteria: root mean square error (RMSE), mean absolute error (MAE), and the scoring function (Score). The mathematical formulations of these metrics are given as (9-11).

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (\hat{y}_i - y_i)^2} \quad (9)$$

$$MAE = \frac{1}{n} \sum_{i=1}^n |\hat{y}_i - y_i| \quad (10)$$

$$Score = \begin{cases} \frac{1}{n} \sum_{i=1}^n e^{-\left(\frac{\hat{y}_i - y_i}{a_1}\right)} - 1, & \hat{y}_i - y_i < 0 \\ \frac{1}{n} \sum_{i=1}^n e^{\left(\frac{\hat{y}_i - y_i}{a_2}\right)} - 1, & \hat{y}_i - y_i \geq 0 \end{cases} \quad (11)$$

Where n represents the number of samples, y_i denotes the actual values, $\bar{y}_i$ is the sample mean, and $\hat{y}_i$ represents the predicted values. The parameters a_1 and a_2 are set to 13 and 10, respectively. Lower root mean square error (RMSE) and mean absolute error (MAE) indicate superior predictive performance, while a

higher Score value signifies greater accuracy in remaining useful life (RUL) estimation.

The comparative results are illustrated in Figure 6, where the blue curve represents the original bearing degradation data, while the red curve denotes the 40% degradation trend predicted by the proposed model. For comparison, the black, green, and gray curves correspond to the predictions of the LSTM, DeepState, and Switching SSM models, respectively. Additionally, the shaded regions indicate the 95% confidence intervals for each model, highlighting the uncertainty in the predictions. To further assess the performance of different models, Fig. 7 presents a comparative analysis of the RMSE, MAE, and Score evaluation metrics.

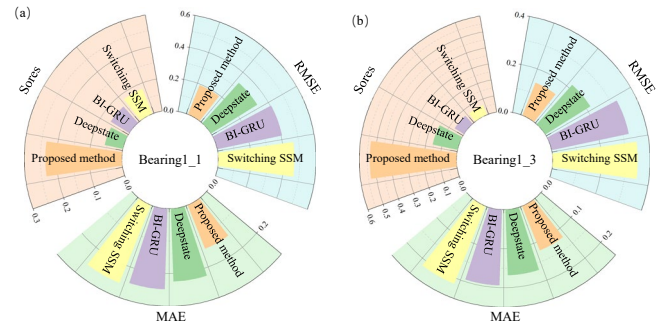


Fig. 7. Prediction error of Bearing1_1 and Bearing1_3

TABLE II. REDICTION ERROR OF BEARING DEGRADATION TREND OBTAINED USING THE FOUR MODELS

Data	Proposed method			BI-GRU			Deepstate			Switching SSM		
	RMSE	MAE	Score	RMSE	MAE	Score	RMSE	MAE	Score	RMSE	MAE	Score
Bearing 1_1	0.193	0.144	0.235	0.334	0.217	0.057	0.405	0.247	0.067	0.46	0.253	0.083
Bearing 1_2	0.296	0.163	0.435	0.292	0.176	0.093	0.776	0.379	0.157	0.455	0.239	0.167
Bearing 1_3	0.191	0.095	0.25	0.138	0.073	0.032	0.542	0.269	0.073	0.291	0.146	0.09
Bearing 2_1	0.459	0.299	0.238	0.991	0.673	0.059	1.194	0.564	0.069	1.048	0.578	0.085
Bearing 2_2	0.311	0.217	0.245	0.988	0.788	0.061	1.089	0.76	0.071	0.996	0.755	0.087
Bearing 2_3	0.482	0.239	0.23	0.541	0.453	0.144	0.861	1.535	0.064	1.295	0.742	0.08
Bearing 3_1	0.273	0.151	0.248	0.327	0.29	0.163	0.749	0.352	0.074	0.45	0.264	0.092
Bearing 3_2	0.473	0.253	0.233	0.509	0.762	0.058	1.779	1.082	0.068	0.92	0.699	0.084
Bearing 3_3	0.234	0.115	0.241	0.905	0.457	0.060	0.598	0.359	0.07	0.546	0.377	0.086
Mean value	0.604	0.318	0.265	0.712	0.504	0.093	1.338	0.702	0.261	0.885	0.507	0.086

Furthermore, the four aforementioned models were applied to nine additional bearing datasets under three different operating conditions from the PHM 2012 dataset. The prediction results are summarized in Table II. Compared to the other three models, the proposed approach demonstrated superior performance. Among the nine predicted bearings, the RMSE and MAE values obtained using the proposed model were consistently lower than those produced by the other models. Specifically, the prediction accuracy improved by at least 10% compared to LSTM, DeepState, and Switching SSM. These results strongly validate the effectiveness and superiority of the proposed model in degradation trend prediction.

IV. CLUSION

To address the limitations of existing prognostic models for rolling bearing degradation trend prediction—namely, their tendency to overlook the heterogeneity between long-term and short-term time series, which leads to reduced performance in practical applications—this study proposes a novel prediction approach that integrates the Mamba framework with a local attention mechanism. The effectiveness of the proposed method is validated using the PHM 2012 dataset. Experimental results demonstrate that the approach effectively leverages both long-range global dependencies and local fine-grained features, thereby capturing the intricate dynamic variations throughout the degradation process more comprehensively. Compared to conventional methods, the proposed approach exhibits significant advantages in early-stage degradation warning and critical phase identification. Moreover, across multiple evaluation metrics, such as root mean square error (RMSE) and prediction residuals, our method consistently achieves higher predictive accuracy and

stability, substantiating the effectiveness of incorporating local attention mechanisms for capturing temporal heterogeneity.

In conclusion, the degradation trend prediction framework proposed in this study, which integrates Mamba-based global modeling with local attention mechanisms, not only addresses the shortcomings of existing prognostic models in handling long-term and short-term time series heterogeneity but also introduces an innovative and effective modeling paradigm for rolling bearing remaining useful life (RUL) prediction

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